

independent events

5. subexperiments independent

$\Leftrightarrow E_1, E_2, \dots, E_n, \dots$ independent
 $\forall E_i$ event of i th subexperiment.

5.2 trials

If each subexperiment has same set of possible outcomes, the subexperiments are called trials.

5.3 independent trials.

1.2. E, F indep $\Leftrightarrow E, F^c$ indep

proof: $\Rightarrow: \because P(E) = P(EF) + P(EF^c)$
 $= P(E)P(F) + P(E)P(F^c)$

$$\therefore P(EF^c) = P(E)(1 - P(F)) = P(E)P(F^c)$$

$\Leftarrow: \because E, F^c$ indep

$\therefore E, (F^c)^c$ indep
 $\therefore E, F$ indep

1.3 E, F indep

$\Leftrightarrow E, F^c$ indep
 $\Leftrightarrow E^c, F$ indep
 $\Leftrightarrow E^c, F^c$ indep

2.2 E, F, G indep

$\Leftrightarrow E, F$ indep } i.e., E, F, G pairwise indep
 E, G indep }
 F, G indep }

E, FG indep.

proof: $\Rightarrow: E, F, G$ indep

$$\Rightarrow P(EFG) = P(E)P(F)P(G) = P(E)P(FG)$$

$\therefore E, FG$ indep

$\Leftarrow: \because E, FG$ indep

$$\therefore P(E(FG)) = P(E)P(FG) = P(E)P(F)P(G)$$

$\therefore E, F, G$ indep.

1. events E, F independent

$$\Leftrightarrow P(EF) = P(E)P(F)$$

2. E, F, G independent

$$\Leftrightarrow P(EFG) = P(E)P(F)P(G)$$

$$P(EF) = P(E)P(F)$$

$$P(EG) = P(E)P(G)$$

$$P(FG) = P(F)P(G)$$

3. events $E_1, E_2, \dots, E_n$ independent

$$\Leftrightarrow \forall \text{ subset } E_1, E_2, \dots, E_r, \quad r \leq n,$$

$$P(E_1 E_2 \dots E_r) = P(E_1)P(E_2) \dots P(E_r)$$

4. an infinite set of events independent

$$\Leftrightarrow \forall \text{ finite subset of these events indep.}$$

3.3.e let $n=4$

E_1, E_2, E_3, E_4 indep

$$\Leftrightarrow \forall E_1, \dots, E_4, \quad 1 \leq 4, \quad P(E_1 \dots E_4) = P(E_1) \dots P(E_4)$$

$$\Leftrightarrow P(E_1 E_2 E_3 E_4) = P(E_1) P(E_2) P(E_3) P(E_4)$$

$$\forall E_1, E_2, E_3, \text{ indep, } i \leq 4$$

3.3 E_1, E_2, E_n indep (i.e., n -wise indep)

(here n -wise is cumulative n -wise)

$$\Leftrightarrow P(E_1 \dots E_n) = P(E_1) P(E_2) \dots P(E_n)$$

$$\forall E_1, E_2, \dots, E_{n-1}, \text{ indep, } i' \leq n$$

(i.e., $E_1, E_2, \dots, E_n$ $(n-1)$ -wise indep)

proof: $\Rightarrow$ ① $\because E_1, E_2, E_n$ indep

$$\therefore P(E_1 E_n) = P(E_1) \dots P(E_n)$$

$$\textcircled{2} \forall E_1, E_2, \dots, E_{n-1}, i' \leq n$$

$$\forall E_1, E_2, \dots, E_{n-1}, \text{ from } \{E_1, E_2, \dots, E_{n-1}\}$$

$$P(E_1 \dots E_{n-1}) = P(E_1) \dots P(E_{n-1}) \quad s \leq n-1$$

$$\therefore E_1, E_2, \dots, E_{n-1} \text{ indep.}$$

2.5.1 let $X = FG$, hold.

2.5.2 let $X = F^c G$.

E, F, G indep

$$\Leftrightarrow E, F^c, G \text{ indep}$$

$$\Leftrightarrow E, F^c, G \text{ pairwise indep}$$

$E, F^c G$ indep

$$\Leftrightarrow E, F, G \text{ pairwise indep}$$

$E, F^c G$ indep

2.5.3 proof:

$$\because X = \begin{cases} FG \\ F G^c \\ F^c G \\ F^c G^c \end{cases} \text{ hold,}$$

$$\therefore X = \begin{cases} (FG)^c \\ (FG)^c \\ (FG)^c \\ (FG)^c \end{cases} \text{ hold.}$$

$$(FG)^c = F^c G$$

□

2.3 E, F, G indep $\Leftrightarrow E^c, F, G$ indep

proof: E, F, G indep

$$\Leftrightarrow E, F, G \text{ pairwise indep}$$

E, F, G indep

$$\Leftrightarrow E^c, F, G \text{ 2-wise indep}$$

E^c, F, G indep

$$\Leftrightarrow E^c, F, G \text{ indep}$$

2.4 E, F, G indep $\Leftrightarrow E', F', G'$ indep

here $E' \in \{E, E^c\}$,

$F' \in \{F, F^c\}$,

$G' \in \{G, G^c\}$

2.5 E, F, G indep

$$\Leftrightarrow E, F, G \text{ 2-wise indep}$$

E, X indep.

here X any event formed from F, G .

$$\Leftarrow \because E_1, E_2, E_3, E_4 \text{ indep}$$

$$\Rightarrow \because E_1, E_2, E_3, E_4 \text{ indep}$$

$$\therefore P(E_1, E_2, E_3, E_4) = P(E_1)P(E_2)P(E_3)P(E_4)$$

$$= P(E_1) \cdots P(E_4)$$

proof:

$$\forall E_1, E_2, E_3, E_4, \text{ indep}, i' \leq 4$$

$$\therefore P(E_1 \cdots E_4) = P(E_1) \cdots P(E_4)$$

$$\therefore P(E_1, E_2, E_3, E_4) = P(E_1)P(E_2)P(E_3)P(E_4)$$

$$\therefore E_1, E_2, E_3, E_4 \text{ indep.}$$

$$\therefore E_1, E_2, E_3, E_4 \text{ indep}$$

$$\therefore P(E_1, E_2, E_3, E_4) = P(E_1)P(E_2)P(E_3)P(E_4)$$

$$\therefore P(E_1 \cdots E_4) = P(E_1) \cdots P(E_4)$$

$$\Rightarrow E_1, E_2, E_3, E_4 \text{ indep.}$$

$$E_1, E_2, E_3, E_4 \text{ 3-wise indep}$$

$$E_1^c, E_2, E_3, E_4 \text{ 3-wise indep}$$

$$\Rightarrow E_1^c, E_2, E_3, E_4 \text{ indep}$$

$$\text{proof: } \Rightarrow P(E_1, E_2, E_3, E_4) = P(E_1) \cdots P(E_4)$$

$$= P(E_1)P(E_2)P(E_3)P(E_4)$$

3.6.e let $n=4$

$$E_1, E_2, E_3, E_4 \text{ indep}$$

$$\Rightarrow F_1, F_2, F_3, F_4 \text{ indep,}$$

here $F_i \in \{E_1, E_1^c\}, \forall i=1, \dots, 4.$

$$E_1, E_2 \text{ indep}$$

$$E_1, E_3, E_4 \text{ indep} \Rightarrow E_1, E_3, E_4 \text{ 2-wise indep}$$

$$E_2, E_3, E_4 \text{ indep} \Rightarrow E_2, E_3, E_4 \text{ 2-wise indep}$$

$$\therefore E_1, E_2, E_3, E_4 \text{ indep.}$$

$$\Leftarrow: \textcircled{1} r=n,$$

$$P(E_1, E_2, \dots, E_n) = P(E_1) \cdots P(E_n)$$

$$\textcircled{2} r=n-1,$$

$$\forall E_1', E_2', \dots, E_{n-1}',$$

$$\therefore \text{indep } \therefore P(E_1', E_{n-1}') = P(E_1') \cdots P(E_{n-1}')$$

$$\textcircled{3} r < n-1,$$

$$\therefore E_1', \dots, E_{r-1}' \text{ is from some } E_1, E_2, \dots, E_{n-1}$$

$$E_1', \dots, E_{r-1}', E_{n-1}' \text{ indep}$$

$$\therefore P(E_1', \dots, E_{r-1}') = P(E_1') \cdots P(E_{r-1}')$$

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3.4.e. let $n=4$

$$E_1, E_2, E_3, E_4 \text{ indep}$$

$$\Leftrightarrow P(E_1, \dots, E_4) = P(E_1) \cdots P(E_4)$$

$$\forall E_1', E_2', E_3', \text{ indep}, i' \leq 4$$

$$\textcircled{1} \Leftrightarrow E_1, E_2, E_3, E_4 \text{ indep}$$

$$\forall E_1', E_2', E_3', \text{ indep}, i' \leq 4$$

proof: (see 3.4)

$$\Rightarrow: P(E_1 E_2 \dots (E_{k+1} E_n))$$

$$= P(E_1 E_n)$$

$$= P(E_1) P(E_n)$$

$$= [P(E_1) P(E_2)] [P(E_{k+1}) P(E_n)]$$

$$= P(E_1 E_2) P(E_{k+1} E_n)$$

$$\Leftarrow P(E_1 E_n)$$

$$= P((E_1 E_2) (E_{k+1} E_n))$$

$$= P(E_1 E_2) P(E_{k+1} E_n)$$

$$= P(E_1) P(E_2) P(E_{k+1}) P(E_n)$$

$$= P(E_1) \dots P(E_n)$$

$$3.9. E_1, E_2, \dots, E_n \text{ indep}$$

$$\Leftrightarrow G_1, G_2, \dots, G_k \text{ indep}$$

$$E_1, E_2, \dots, E_n \text{ (n)-wise indep.}$$

$$E_1, E_2, E_{k+1} \text{ indep}$$

$$\Leftrightarrow E_1, E_2 E_{k+1} \text{ indep}$$

$$E_1, E_{k+1} \text{ k-wise indep}$$

$$\Leftrightarrow E_1^c, E_2 E_{k+1} \text{ indep}$$

$$E_1^c, E_2, E_{k+1} \text{ k-wise indep}$$

$$\Leftrightarrow E_1^c, E_2, E_{k+1} \text{ indep.}$$

$$3.6. E_1, E_2, \dots, E_n \text{ indep}$$

$$\Leftrightarrow F_1, F_2, \dots, F_n \text{ indep}$$

$$\text{here } F_i \in \{E_i, E_i^c\}$$

$$3.7. E_1, E_2, \dots, E_n \text{ k-wise indep}$$

$$\Leftrightarrow F_1, F_2, \dots, F_n \text{ k-wise indep}$$

$$\text{here } F_i \in \{E_i, E_i^c\}, 2 \leq k \leq n$$

$$3.8. E_1, E_2, E_n \text{ indep}$$

$$\Leftrightarrow E_1 \dots E_1, E_{(k+1)} \dots E_n \text{ indep}$$

$$E_1, E_2, E_n \text{ (n)-wise indep.}, k \leq n.$$

$$3.4. E_1, E_2, \dots, E_n \text{ indep}$$

$$\Leftrightarrow P(E_1 \dots E_n) = P(E_1) \dots P(E_n)$$

$$E_1, E_2, \dots, E_n \text{ (n)-wise indep}$$

$$\Leftrightarrow E_1, E_2 \dots E_n \text{ indep}$$

$$E_1, E_2, \dots, E_n \text{ (n)-wise indep}$$

$$\text{proof: } \Rightarrow: P(E_1 (E_2 E_n)) = P(E_1) P(E_2) P(E_n)$$

$$= P(E_1) P(E_2 \dots E_n)$$

$$\therefore E_1, E_2 E_n \text{ indep}$$

$$\Leftarrow: E_1, E_2 E_n \text{ indep}$$

$$\therefore P(E_1 (E_2 E_n)) = P(E_1) P(E_2) P(E_n)$$

$$= P(E_1) P(E_2) P(E_n)$$

$$3.5. E_1, E_2, E_n \text{ indep} \Leftrightarrow E_1^c, E_2, E_n \text{ indep}$$

proof: for $n=2, 3, 4$, hold

if $n=k$, hold,

then for $n=k+1$,

here $2 \leq k \leq n$,

$$G_1 = F_{i_1} F_{i_2} \dots F_{i_{p_1}}$$

$$G_2 = F_{i_1} F_{i_2} \dots F_{i_{p_2}}$$

$$G_k = F_{i_1} F_{i_2} \dots F_{i_{p_k}}$$

$$p_1 + p_2 + \dots + p_k = n$$

$i_1, i_2, i_{p_1}, i_1, i_{p_2}, \dots, i_{p_k}$ is permutation of $1, 2, \dots, n$.

$$F_j \in \{E_j, E_j^c\}, \forall j \in \{1, \dots, n\}$$

proof: E_1, E_2, E_n indep

$$\Leftrightarrow F_1, F_2, F_n \text{ indep}$$

$$\Leftrightarrow G_1, G_2, G_k \text{ indep}$$

$$F_1, F_2, F_n \text{ (n-1)-wise indep (from 3.9)}$$

$$\Leftrightarrow G_1, G_2, G_k \text{ indep}$$

$$E_1, E_2, E_n \text{ (n-1)-wise indep.} \quad \square$$

$$= P(E_{i_1} E_{i_2} \dots E_{i_j}) P(E_{i_{j+1}} E_{i_{j+2}} \dots E_{i_n})$$

$$= P(G_1) \dots P(G_j)$$

② $\forall j-1$ events $G_1, \dots, G_{(j-1)}$ from

$$\{G_1, G_2, \dots, G_j\}$$

" E_1, E_2, E_n indep

" $E_{i_1}, E_{i_2}, E_{i_{p_1}}, E_{i_{(j-1)+1}}, E_{i_{(j-1)+2}}, \dots, E_{i_{(j-1)+p_{(j-1)}}}$ indep

" $G_1, \dots, G_{(j-1)}$ indep (" $k=j-1$ true) proof:

" G_1, G_2, G_j indep

$$\Leftarrow P(E_1 E_n) = P(G_1 G_j) = P(G_1) P(G_j) = P(E_1) P(E_n) \text{ then, for } k=j,$$

" $k=j$ true

$$\Rightarrow \textcircled{1} P(G_1 G_j)$$

$\square$

3.10 E_1, E_2, E_n indep

$$\Leftrightarrow G_1, G_2, G_k \text{ indep,}$$

E_1, E_2, E_n (n-1)-wise indep

here $2 \leq k \leq n$,

$$G_1 = E_{i_1} E_{i_2} \dots E_{i_{p_1}},$$

$$G_2 = E_{i_1} E_{i_2} \dots E_{i_{p_2}},$$

$$G_k = E_{i_1} E_{i_2} \dots E_{i_{p_k}}$$

$$p_1 + p_2 + \dots + p_k = n$$

$i_1, i_2, i_{p_1}, i_1, i_{p_2}, \dots, i_{p_k}$ is permutation of $1, 2, \dots, n$.

for $k=2$, true (from 3.8)

if $k=j-1$, true.

$$= P(E_{i_1} E_{i_2} \dots E_{i_{j-1}}) (E_{i_j} E_{i_{j+1}} \dots E_{i_n})$$

$$= P(E_1) P(E_n)$$

$$= [P(E_{i_1}) P(E_{i_2} \dots E_{i_{j-1}})] [P(E_{i_j}) P(E_{i_{j+1}} \dots E_{i_n})]$$

$$F_i^{(j)} \in \{G_i^{(j)}, G_i^{(j)c}\}, \forall i \in \{1, \dots, k_j\}$$

$$G_1^{(H)} = F_{i_1}^{(j)} \dots F_{i_{p_{(H)}}}^{(j)}$$

$$\dots$$

$$G_{k_H}^{(H)} = F_{i_{k_H}}^{(j)} \dots F_{i_{p_{(H)}}}^{(j)}$$

$$p_1^{(H)} + p_2^{(H)} + \dots + p_{k_H}^{(H)} = k_j,$$

$$i_1^{(H)}, i_{12}^{(H)}, \dots, i_{k_H}^{(H)} p_{k_H}^{(H)}$$

permutation of $1, \dots, k_j$

proof: $G_1^{(0)}, G_{k_0}^{(0)}$ indep

$$\Leftrightarrow G_1^{(1)}, G_{k_1}^{(1)} \text{ indep}$$

$$G_1^{(0)}, G_{k_0}^{(0)} \text{ (k-1)-wise indep (from 3.10)}$$

$$\Leftrightarrow G_1^{(2)}, G_{k_2}^{(2)} \text{ indep}$$

$$G_1^{(1)}, G_{k_1}^{(1)} \text{ (k-1)-wise indep}$$

$$G_1^{(0)}, G_{k_0}^{(0)} \text{ (k-1)-wise indep of } 1, \dots, k_0$$

$$\Leftrightarrow G_1^{(2)}, G_2^{(2)}, G_{k_2}^{(2)} \text{ indep,}$$

$$G_1^{(0)}, G_2^{(0)}, G_{k_0}^{(0)} \text{ (k-1)-wise indep.}$$

$$\Leftrightarrow G_1^{(1)}, G_2^{(1)}, G_{k_2}^{(1)} \text{ indep,}$$

$$G_1^{(0)}, G_2^{(0)}, G_{k_0}^{(0)} \text{ (k-1)-wise indep.}$$

$$\forall j = 0, 1, 2, \dots$$

$$k_0 > k_1 > k_2 > \dots > k_j > 2$$

$$\text{here } F_i^{(0)} \in \{G_1^{(0)}, G_2^{(0)}, \dots, G_{k_0}^{(0)}\}, \forall i \in \{1, \dots, k_0\}$$

$$G_1^{(1)} = F_{i_1}^{(0)} \dots F_{i_{p_1}}^{(0)},$$

$$G_2^{(1)} = F_{i_2}^{(0)} \dots F_{i_{p_2}}^{(0)},$$

$$G_{k_1}^{(1)} = F_{i_{k_1}}^{(0)} \dots F_{i_{p_{k_1}}}^{(0)}$$

$$p_1^{(1)} + p_2^{(1)} + \dots + p_{k_1}^{(1)} = k_0$$

$$i_1^{(1)}, i_{12}^{(1)}, \dots, i_{k_1}^{(1)} p_{k_1}^{(1)} \text{ is permutation}$$

$$G_1^{(0)}, G_{k_0}^{(0)} \text{ (k-1)-wise indep of } 1, \dots, k_0$$

$$3.11 \quad E_1, E_2, E_n \text{ (n-1)-wise indep}$$

$$\Rightarrow G_1, G_2, G_k \text{ (k-1)-wise indep.}$$

$$\forall i = 0, 1, \dots, k-2.$$

$$(G_1, G_2, G_k \text{ from 3.10, } 2 \leq k \leq n)$$

proof: $\forall k$ -events $G_1, G_2, \dots, G_{k-1}$

from $\{G_1, G_2, G_k\}$

$$\therefore E_1, E_2, E_n \text{ (n-1)-wise indep}$$

$$\Leftrightarrow p_1 + p_2 + \dots + p_{k-1} \leq n-1$$

$$\therefore E_{i_1}, E_{i_2}, \dots, E_{i_{p_1}}$$

$$E_{i_2}, E_{i_3}, \dots, E_{i_{p_2}}$$

$$E_{i_{k-1}}, E_{i_{k-1}+1}, \dots, E_{i_{k-1}+p_{k-1}} \text{ indep.}$$

$$\therefore G_1, G_2, \dots, G_{k-1} \text{ indep (from 3.10)}$$

□

$$3.12 \quad G_1^{(0)}, G_2^{(0)}, G_{k_2}^{(0)} \text{ indep}$$

$$\Leftrightarrow G_1^{(1)}, G_2^{(1)}, G_{k_2}^{(1)} \text{ indep, } G_1^{(0)}, G_2^{(0)}, G_{k_0}^{(0)} \text{ (k-1)-wise indep.}$$

∴ $H_1 = H'_1$, H'_1 is formed from 3.1.3

E_{i_1}, E_{i_p} , using only $\wedge, c$.

Similarly, $H_2 = H'_2, \dots, H_k = H'_k$.

Let $G_1^{(0)} = E_1, G_2^{(0)} = E_2, G_{k_0}^{(0)} = E_n$,

$\Rightarrow \exists j$, s.t., $G_1^{(j)} = H'_1, G_2^{(j)} = H'_2, G_{k_j}^{(j)} = H'_k$,

($G_1^{(0)}, \dots, G_{k_j}^{(j)}$ from 3.1.2).

∴ $G_1^{(0)}, G_2^{(0)}, G_{k_0}^{(0)}$ indep

$\Leftrightarrow G_1^{(j)}, G_2^{(j)}, G_{k_j}^{(j)}$ indep,

$G_1^{(0)}, G_2^{(0)}, G_{k_0}^{(0)}$ (k_0-1)-wise indep.

$\Leftrightarrow H_1, H_2, H_k$ indep,

E_1, E_2, E_n ($n-1$)-wise indep.

□.

using $\wedge, \vee, c$.

∴ De Morgan's laws:

$$(\cup E_i)^c = \cap E_i^c$$

E_1, E_2, E_n indep

$\Leftrightarrow H_1, H_2, H_k$ indep,

E_1, E_2, E_n ($n-1$)-wise indep

here $2 \leq k \leq n$

H_1 any event formed from

$E_{i_1}, E_{i_2}, \dots, E_{i_p}$

H_2 formed from $E_{i_2}, E_{i_2}, \dots, E_{i_{p_2}}$

H_k formed from $E_{i_{k_1}}, E_{i_{k_2}}, \dots, E_{i_{k_{p_k}}}$

(note each E is used only once by some H)

$$p_1 + p_2 + \dots + p_k = n$$

$i_1, i_2, i_{p_1}, i_2, i_1, \dots, i_{k_{p_k}}$ is permutation of $1, 2, \dots, n$.

$G_1^{(0)}, G_2^{(0)}, G_{k_0}^{(0)}$ indep

$\Leftrightarrow G_1^{(j)}, G_2^{(j)}, G_{k_j}^{(j)}$ indep

E_1, E_2, E_n ($n-1$)-wise indep.

□

$\Leftrightarrow G_1^{(j)}, G_{k_j}^{(j)}$ indep

$G_1^{(H)}, G_{k_j}^{(H)}$ (k_j-1)-wise indep

$\dots$

$G_1^{(1)}, G_{k_1}^{(1)}$ (k_1-1)-wise indep

$G_1^{(0)}, G_{k_0}^{(0)}$ (k_0-1)-wise indep

$\dots G_1^{(j)}, G_{k_j}^{(j)}$ (k_j-1)-wise indep

$\Rightarrow G_1^{(H)}, G_{k_j}^{(H)}$ (k_j-1)-wise indep

(from 3.1.1)

$\forall j=0, 1, 2, \dots$